

RAG over a rulebook

Cite the exact clause – or refuse

Offline question answering over four engineering rulebooks: every answer carries an exact (file, §) address – or says the question isn't covered. A case study in evaluation discipline; corpus and answers in Russian.

0 / 680

answers without an address

72%

recall@20 on a clean holdout

58

predictions written before the runs

0.666

nDCG@10 against a 0.665 anchor

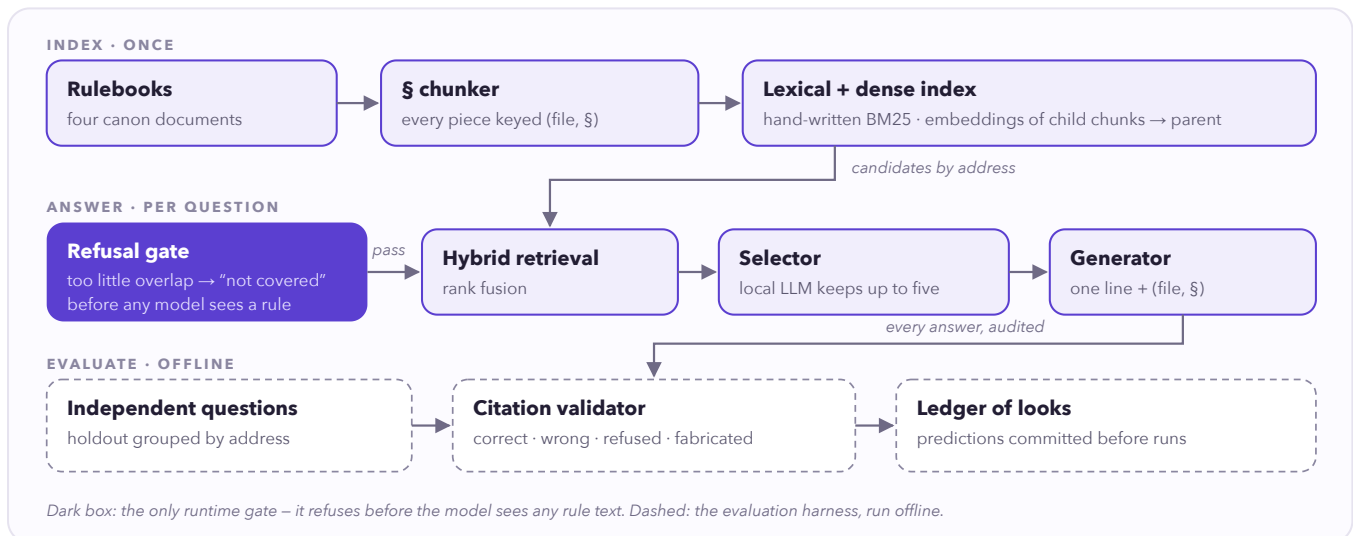
THE PROBLEM

An answer is only as good as the clause behind it, and the clause must be checkable. Almost half of the § numbers exist in more than one document, so only a (file, §) pair is a valid citation. A small model given any context never refuses on its own, and questions written by the author overstated retrieval by 28 points.

WHAT IT DOES

- Indexes addressed § sections twice – lexical and dense
- Refuses before generation when a question barely overlaps the rules
- Fuses both rankings; a local model answers in one line with an address
- Measures itself on independent questions, logging every holdout look

ARCHITECTURE



HOW IT FLOWS

- 1 Sections keep their (file, §) address; small child chunks point back to the parent.
- 2 A question meets the refusal gate first – too little overlap, and it stops there.
- 3 Rankings are fused; a local model keeps up to five fragments and answers in one line.
- 4 Offline, a validator sorts every answer: correct, wrong, refused, unaddressed or fabricated.

WORTH KNOWING

A hand-written BM25 matched a published anchor.

nDCG@10 0.666 against 0.665 on BEIR/SciFact, 300 queries – with the pass bar committed before the run.

The gate reveals the loss; it doesn't cause it.

Without it, 0 of 7 false refusals became useful answers, and 3 of 3 correct refusals became confident wrong ones.

Author-written questions flattered retrieval by 28 points.

The first run on 394 independent questions switched the retriever from lexical to hybrid.

DECISIONS THAT SHAPE IT

Hybrid retrieval over pure lexical or pure vector 01

Because on 314 independent questions hybrid reached 76% recall@20 against 61% lexical and 71% dense; a pre-registered paired test on the clean holdout confirmed it – 72% against 57%, $p = 0.0018$.

Cost: fusion discards score magnitude, so refusal needs its own gate.

The § section as the unit, (file, §) as the key 02

Because almost half of the § numbers collide across documents. Parent-document retrieval keeps the address and lifted dense top-5 from 6% to 61%.

Cost: uneven sections, and special cases for chapters without text.

Citations by structure, not by prompt wording 03

Because with fragment headers 8 of 8 answers carried a file-qualified address; without them, 1 of 8 – whatever the prompt said.

Cost: a valid citation is not a correct one, so correctness is measured separately.

A lexical refusal gate, not the model’s judgment 04

Because with empty context the model refused 210 of 210; given any context, it refused 0 of 102.

Cost: the gate measures word overlap, so short real questions are refused more often.

A ledger of looks; predictions before runs 05

Because the first architecture choice was made with the holdout inside the denominator. A pre-registered rerun put the cost at about 4 points – and the ranking held.

Cost: slower loops, and negatives get published.

Calibrate the ruler on someone else’s benchmark 06

Because internal numbers can’t validate the instrument itself. BM25 matched the BEIR anchor within 0.001 – and the log calls the third decimal “partly luck”.

Cost: it validates only the lexical channel, in English.

HOW IT IS ENGINEERED



Limits are stated above the strengths. The README opens with the honest table: exact (file, §) accuracy is 50% on an independent holdout – each number with its own configuration and population, so no headline figure can hide behind a friendlier one.

WHERE IT TRANSFERS

Wherever an answer is only as good as the clause behind it: internal policies and HR handbooks, compliance and regulatory manuals, security standards, SOPs, support knowledge bases. What carries over is the architecture, not the model – index at clause level, make (document, clause) the citation key, enforce citations by structure and give refusal its own path – and the evaluation harness carries over unchanged, so a new corpus means re-measuring, not re-arguing.

STACK

Core Python standard library only – BM25 with a Russian stemmer, rank fusion, parent-document retrieval, written by hand · **Models** Ollama: Qwen3 as selector and generator, bge-m3 embeddings · **Statistics** Wilson intervals, cluster bootstrap, exact McNemar, minimum detectable effect